Which Factors Contribute to the Accuracy and Efficiency of Horse Racing Betting Markets?

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ABSTRACT

Prediction markets have become increasingly popular over the last few years with the advent of new cryptocurrency services such as Polymarket and Kalshi. A prediction market allows users to buy and sell assets with values that depend on a future outcome. An example of a prediction market with a long history is that of the racetrack. Markets at the racetrack are extremely regular, with a lot of repeated similar events making them valuable research targets. The structure of the markets is not dissimilar to those of traditional financial markets, enabling comparisons to be drawn. This report seeks to identify whether racing markets can be considered accurate or efficient. Accuracy and efficiency are two key metrics indicating how and when a prediction market is useful. It is typical for investment firms to create funds of stocks grouped by certain categories such as industry or market location in order to meet different investment requirements of risk. Different distinctions of horse races are likewise developed in this report and examined. A novel approach is taken with the decomposition of the market data into multiple market classification schemes. The market examined utilises a continuous double auctions, rather than the parimutuel market typically examined in the prevailing research. It is found that, in general, the UK horse racing exchange Betfair can be considered accurate in odds generation. Certain factors affect this accuracy, such as the race number or the number of runners. No strong evidence is found to imply a lack of market efficiency for the entire racing market or any of the groups.

Acronyms

BHA — British Horseracing Authority
BSP — Betfair Starting Price
CDA — Continuous Double Auction
CLOB — Central Limit Order Book
DPM — Dynamic Parimutuel Market
HKJC — Hong Kong Jockey Club
IP — In-play (after the off)
PNL — Profit and Loss
PP — Pre-play (before the off)
WAP — Weighted Average Price

Glossary

Arbitrage — A risk-free opportunity to make money

Back — A bet that something will happen

Daily Double — A bet on the winner of two consecutive races

Each-way — A combination of a win and place bet

Exacta — A bet on two horses which must come first and second

Lay — A bet that something will not happen

Off — The start of the race

Options Contract — A right but not an obligation to buy or sell a security at a given date.

Overround — The total implied probability of all events in a market

Pandas — Python library for data manipulation

Parlay — A single bet which links multiple consecutive wagers

Place — Come first or second in a race

Prediction Market — A market that facilitates bets on the outcomes of future events

Python — Programming Language

Show — Come first, second or third in a race

Tipster — Professional who publishes betting guides

1. Introduction

A recent surge in the popularity of prediction markets has brought them to the forefront of public discussion. Services such as Polymarket and Kalshi are offering the opportunity to bet on many world events. There has been research to suggest that prediction markets are more accurate than other forecasting techniques including polls (Berg, 2008). Horse racing markets are also fundamentally prediction markets, and have a long history of widespread use. As a prediction market, horse races provide good data since there are so many and the approximate date and time is known before hand. It is easier to determine their accuracy in this context as there are many repeated similar markets, unlike predictions for political events which can be very far apart.

Racing markets can be considered largely analogous to traditional financial markets, with the only real difference being social convention (Snyder, 1978). Wagers can be considered similar to binary options contracts, where there are two different payoff values dependent on the result of an external event. The difference between options and wagers here is that the value of an option is dependent on the value of an underlying asset at a future date. Because of this, options are priced using the Black-Scholes model (1973) where it can be shown that buying a combination of the underlying asset and a risk-free investment can produce a payoff equivalent to the option. If not priced this way, an arbitrage is apparent and an inefficiency in the market occurs. Without an underlying asset to hedge against, wagers must be priced differently. Bets are fundamentally different from stocks as the lifetime of a bet is finite and the lifetime of a stock is not (Thaler, 1988).

In wider financial markets, it is typical for index funds to be selected from a certain subsection of securities. For example the S&P 500 tracks the top 500 stocks in the United States and the FTSE 100 tracks the top 100 companies on the London Stock Exchange. Both of these are examples of funds limited to investments within a certain region. Despite this, little research has been done on how prediction markets can be classified in similar ways for bettors to make informed decisions on selected groups of races only.

1.1. Motivation

I was motivated to research this area due to the recent popularity of prediction markets as a tool for forecasting. As well as this, the regulation surrounding prediction markets is currently heavy within the UK. This is because they fall under the jurisdiction of the Gambling Commission. Many of the newly popular prediction market sites are currently illegal to use within the UK, but established exchanges such as Betfair are completely legal.

As prediction markets are considered gambling, winnings within the UK are not taxed. This means that an equivalent strategy within a gambling market and the stock market may prove ultimately more profitable in the gambling market as the only fees to be paid are commission to the exchange. In the case of Betfair, commission can be even less than just 2% of successful bets.

I want to understand how different classifications of races can give us new insights into the markets. These insights can be used to aid in the identification of good practices for eliciting forecasting odds, or for the production of profit through betting strategies. Additionally, this research may be applicable outside of the horse racing context, especially in general prediction markets, but also financial markets as a whole.

1.2. Scope

This project explores how the classification of horse races can affect the accuracy and efficiency of the market. The classifications are both external factors and internal market factors. Existing research is used to determine the best practices for identifying accuracy and efficiency. A sample database of horse races will be compiled and tested against to obtain results.

1.3. Aims

This project aims to explore the relevance and impact of external and market factors on the accuracy and efficiency of horse racing markets. Horse races can be classified by various external factors such as race time, location and distance. Internal classifications involve market information such as volume or overround. Similarly, traditional financial instruments can be classified by external factors like their industry or location of operations. They also feature many of the same internal market factors including volume.

1.4. Objectives

To accomplish my aims I will:

- Perform a systematic review of existing research to determine existing standards for market accuracy and efficiency. Identify whether racing markets are generally considered to be accurate or efficient.
- Obtain an appropriate dataset for research.
- Determine the accuracy of my dataset.
- Determine the efficiency of my dataset.
- Determine several classifications of horse races to compare.
- Determine the relative accuracy of the classifications.
- Determine the relative efficiency of each of the classifications.
- Attempt to exploit any inefficiencies for profit.

2. Background

2.1. Bookmaking

2.1.1. Fixed-Price Odds

A fixed-price bookmaker will quote prices (odds) at which they are willing to sell bets. If the bet pays out, the bettor receives the return originally quoted to them. The bettor is given a guarantee of their winnings before placing the bet. It is important to note that it is unlikely a bookmaker will allow a bettor to ‘‘lay’’ a bet, that is to bet that something *will not* happen. Bookmakers make their money by appropriately setting odds such that in all cases (or at least over a period of time), they will ensure a profit. This is done by changing odds to account for the amount of money bet already (balancing the books) and by providing unfair odds that do not sum to one.

There are multiple different ways to represent odds prices. Throughout this report, I will use decimal odds. Decimal odds are given as real numbers greater than one which represent the total payout for a winning bet. For example if a bettor bets £100 dollars at odds of 2, their

winning payout would be £200 - a profit of £100 or 100%. The payout for all losing bets is 0 (a loss of all money bet). Assuming that the expected value of a wager is zero (no profit or loss), the implied probability of winning is given by $\frac{1}{odds}$.

2.1.2. Parimutuel Odds

Parimutuel betting (commonly known as a 'tote' in the UK) is a system in which money for all bets is placed into a pool and then distributed to winning bettors. This means that bettors are not given an exact payout quote at the time of purchasing their bet, but they may be given an estimate. Tote operators make money by taking a certain percentage of the pool for themselves, typically called the ''track take''. These types of bets are typically facilitated by the track operator themselves, such as the HKJC (Hong Kong Jockey Club, n.d.).

2.1.3. Betfair Exchange

Betfair is a UK based company which operates a betting exchange service. A betting exchange is different from a typical bookmaker in that instead of the operator quoting odds prices or offering a tote service, it allows users to form bets with one another. Betfair does this using a CLOB (Central Limit Order Book), allowing users to either back (long) a bet or lay (short) one. Within the context of a horse race, this means that the win market (where bets are paid out only if the horse places first) features a different asset for every selection (runner) in the race. They can be thought of as largely analogous to options contracts, where the layer agrees to pay the backer a certain sum of money dependent on the outcome of the race.

A unique feature of the Betfair racing market is its ''starting price'' betting. Bettors who use the starting price option have their bets matched into the market at the off (when the race begins). They are also able to set minimum odds to back or maximum odds to lay. The price determination maximises volume and gives backers and layers the same price. In order to do that, existing positions within the CLOB may be matched against as well. Starting price bets aren't guaranteed to be matched if the odds limit is at or above the starting price (Betfair, n.d.).

2.2. Literature Review

2.2.1. Accuracy of Horse Racing Markets

2.2.1.1. Measurement of Forecast Accuracy

One of the earliest methods for scoring the accuracy of a forecast is provided by Brier (1950). The Brier score is determined as a result of many predictions using the mean squared difference between the estimated probability and the actual outcome. Formally:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. In this system, a perfect forecast is given a score of zero. This can only be achieved if the forecast guesses with 100% certainty every time. Ranked Probability Score (Epstein, 1969) is an extension of the Brier score to support multiple outcomes. Ranked Probability Score with binary outcomes can be shown to be equivalent to the Brier score (Weigel, 2007). Brier score will be the most useful in this report, as in (most) betting markets there are

only two possible outcomes: a winning bet or a losing bet.

Good (1952) developed a method for scoring estimates of probabilities now known as log-loss. This method has since become popular as a scoring mechanism for machine learning models as it highly punishes overconfidence in incorrect predictions. For a binary outcome:

$$LL = -\frac{1}{N}\sum_{t=1}^{N} o_t \log(f_t) + (1 - o_t)\log(1 - f_t)$$

where f_t is the forecast probability, o_t is the actual outcome (0 if did not happen or 1 if did) and N is the number of forecasts. Log-loss has become incredibly popular as a cost function (function that scores prediction accuracy) for machine learning. Machine learning textbooks (Goodfellow, 2016; Murphy, 2022) share that log-loss should be considered the standard cost function. Since log-loss easily supports binary outcomes, it can be applied in addition to Brier score as a measurement of accuracy for betting predictions.Both Brier scores and log-loss will be useful in ranking the accuracy of markets. The primary difference between them is how harshly they rank confident incorrect guesses.

Skill scores are a way to measure forecast skill relative to a given baseline (Murphy, 1988). These can be applied to both the Brier score and log-loss.

$$SS = 1 - \frac{score}{score_{reference}}$$

A skill score of zero indicates that the forecast is just as good as the baseline forecast. Conversely to the normal Brier score and log-loss, a higher skill score is an indication of better performance. Negative skill scores are worse than the reference and positive skill scores are better than the reference.

Wunderlich and Memmert (2020) demonstrated that the betting returns of a strategy are not necessarily indicative of accuracy in a forecasting model, despite demonstrating profitability. Instead, there is a difference in a model forecasting accurately and profitably.

2.2.1.2. Use of Markets for Forecasting

Current research shows prediction markets to provide accurate estimates of outcomes for elections over both long and short time frames (Berg, 2008). Rothschild (2009) showed that, when used appropriately, prediction markets can perform significantly better than the very best polls. Elections are just one example, but prediction markets have been used to great success across many different domains of forecasting. One study found prediction markets to outperform a survey when used to predict the reproducibility of studies published in psychology journals (Dreber, 2015).

Because of the demonstrated favourite-longshot bias (see more below) in horse racing, many studies attempt to convert odds given by bookmakers or markets into ‘‘real’’ probabilities. Štrumbelj (2014) used Shin’s (1993) model to create more accurate forecasts than those created through odds normalisation or regression. These methods are criticised by Clarke et al. (2017) for potentially producing negative probabilities and in the case of normalisation, not accounting for the favourite-longshot bias. They share that an alternative power method instead performs as well as Shin’s, or better. The necessity of these models in the first place demonstrates the lack of accuracy present within the markets. If the given odds can be altered using some function to provide forecasts of greater accuracy, then the market obviously isn’t producing the best possible

forecasts inherently.

Strijbis and Arnesen (2019) find that the setup of the market influences accuracy the most, while features of the event and the market's participants have an insignificant impact. Market setup incorporates factors such as liquidity and the market mechanism. Prediction markets require the right incentives in order to produce the most accurate and up to date forecasts (Chen, 2010). Markets will also suffer if a group of insiders are able to influence the outcome. If the goal of the market is to produce the best forecast, then it must incentivise immediate truth-telling instead of deception. A typical parimutuel racing market fundamentally does not meet this requirement. This is because market participants are encouraged to wait until the very last moment to bet, so as to hide their beliefs about the probability of a win. A continuous double auction (the primary mechanism of the Betfair Exchange) encourages price discovery over time but may still incentivise participants to lie about their true valuation. Since the price of a parimutuel market is not set until the market closes, there is no incentive for the market to track changes in information unlike a CDA.

Pennock (2004) suggests an alternative ‘‘dynamic pari-mutuel market’’ (DPM) which functions similarly to typical parimutuel betting. A DPM allows investors to buy and sell bets at a changing rate determined by the market, giving them an opportunity to hedge or sell their bets early.

2.2.1.3. Favourite-Longshot Bias

The favourite longshot-bias has been demonstrated within racing markets as far back as 1949 (Griffith, 1949). This is a phenomenon in which bettors tend to prefer to bet on outlier horses, as opposed to the safer options. In turn, this moves the odds given by the market as favourite horses are given worse implied probabilities of winning and longshots are given better implied probabilities of winning. Other than the long and short-odded horses, Griffith found that the odds for horses were on average accurate.

Some studies have attempted to determine why this is the case. Weitzman (1965) used utility theory to derive a utility curve for one bettor and found it to be similar to the theoretical one proposed by Markowitz (1952). To follow this study, Ali (1977) observed over 20,000 races and found the favourite-longshot bias to be present. Ali creates a wealth utility function and determines that the average bettor values risk, thereby causing the discrepancy in prices.Subsequent research has also conjectured that bettors have a preference for risk at the racetrack. Kanto et al. (1992) found this to be the case using a data sample from Finland.

Golec and Tamarkin (1998) suggested an alternative explanation for the favourite-longshot bias. They claimed that it was possible that bettors could actually be risk-averse and instead have a preference for skewness. Skewness here is the asymmetry of betting returns about the mean. To back this up they examined popular betting strategies and determined that they were better aligned with a preference for skewness rather than risk. Another theory is put forward by Snowberg and Wolfers (2010) which states that the bias can be accounted for by the misperceptions of risk by the bettors.

The bias has also been shown to not be universal. In contrast to the evidence found in North American markets, Busche and Hall (1988) found that bettors at the Hong Kong Jockey Club do not exhibit the same apparent risk-loving behaviour. They posit that the larger betting volume in Hong Kong may influence the bias.

2.2.2. Efficiency of Horse Racing Markets

2.2.2.1. Measurement of Market Efficiency

Fama (1970) set out three tests of efficiency for markets depending on how rigorously they can be described as efficient. An efficient market is one that is said to fully incorporate all the available information into the prices. This means that, for an efficient market, it should be impossible to create a profit generating strategy. For a market to be *weak form* efficient, the information set is constrained to that of historical price data. The *semi-strong form* tests also incorporate information widely available to the general public. In the *strong form* tests, the insight of investors with insider knowledge is also considered.

A significant amount of research into the study of efficiency in racing markets is specifically focussed on parimutuel markets. This is the most common market for racing in the United States as well as Hong Kong. Comparatively little exists for markets using a CDA such as Betfair. This makes this report particularly interesting as a way to compare parimutuel markets with a CDA. It also means that this work has greater applicability to external financial markets, which typically use a form of CDA as their market mechanism.

2.2.2.2. Relationship of Exotic Bets

A lack of market efficiency can be demonstrated by comparing the prices of exotic betting options with equivalently produced parlays. Exotic bets involve betting on a combination of different outcomes for different horses within one wager. Popular examples include the daily double, in which the first two winning horses of the day must be predicted correctly, and an exacta, where the first two horses of a race must be predicted in order. A parlay is similarly a wager of multiple outcomes, constructed from multiple separate bets with the winnings from earlier bets accumulated into future ones.

Certain exotic bets can be shown to have an equivalent payoff function to an appropriately constructed parlay. For the market to be considered efficient, both the exotic bet and the parlay must have an equivalent price. This difference in a typical financial market may be considered an arbitrage opportunity as one would be able to buy at the low price and sell at the high price, guaranteeing a profit. Within parimutuel markets however this is not possible, as bets cannot be bought and sold at different prices. This is fundamentally similar to established options pricing theory. The payoff of an option can be shown to be replicated by purchasing a quantity of the underlying asset and a risk free asset (Black, 1973; Cox, 1979).

Ali's (1979) analysis of 1,089 racing days uses this principle to compare the prices of daily doubles and their equivalently constructed parlays. In this analysis, Ali finds that the efficient market hypothesis could possibly hold for racing markets as the difference in price between the exotic bets and the parlays is not statistically significant.In a different study however, Asch and Quandt (1987) find that for both exacta and daily double bets, their payoffs are not the same as equivalent bets on individual horses, thus demonstrating market inefficiency.

It is uncommon for betting exchanges to offer any betting options for horse races outside of win, place and each-way. This means that a comparison of exotic bets and parlays is not possible. While it may be possible to compare the prices of the win and place bets to the each-way bets, the limited volume of non-win markets makes this difficult.

2.2.2.3. Betting Strategies

The most obvious way to demonstrate a lack of market efficiency is to produce a simple, profitable betting strategy.

In a study of 1,730 races in American parimutuel win markets, Snyder (1978) showed that because of the favourite-longshot bias, prices were indeed more favourable for the favourite horses, but the track take was too significant for a strategy of favourite betting to be profitable. This indicates some level of *weak form* efficiency, however without the track take present (and betting habits remaining the same), it would certainly be impossible to describe the market as informationally efficient. Snyder also demonstrated that this bias exists among betting experts (or ''tipsters'') which makes it seem unlikely that the market is *strong form* efficient.

Hausch et al. (1981) set about creating a technical betting system for place and show markets. Their strategy only used historical pricing data and so attempted to demonstrate a lack of *weak form* efficiency.They argued that their system successfully exploited a market inefficiency and was able to generate a consistent profit. To do this, they derived estimated place and show probabilities from the implied win probabilities given by the win market using Harville's (1973) method. The win market is assumed to be informationally efficient. This allows them to profit through mispricing of show and place odds relative to the win market. With a follow up paper, Hausch and Ziemba (1985) greatly improved the system making it more profitable and applicable to any racetrack. They also demonstrated how more people using the system would ultimately cause the market to become efficient.

3. Specification & Design

3.1. Requirements

First of all, a large, high quality data set needs to be sourced. This is especially important, as the data will be decomposed into smaller groups and so a large initial dataset means there will still be sufficient data points to analyse after this. The software needs to support flexible classifications of races so that they can be quickly altered, adjusted, added or removed as required. This also makes the software easier to maintain and manage. With the data, metrics determining the accuracy of the market need to be calculated and strategies need to be back tested.

3.2. Data Selection

I considered three different data sources for use in this project. The first source is the officially provided Betfair historical pricing data. This information is available at a free tier for all months the data is offered. The data is structured like exchange data, documenting price changes over time. For my purposes, this is beyond what I need as I am only interested in the BSP (Betfair Starting Price), and not the price evolution.

Data from the Hong Kong Jockey Club was also considered. After obtaining a dataset by scraping the publicly available results data on their website, I determined it to be insufficient in size as there were only 2,000 races available.

The dataset I ultimately chose was from Betfair, but not through their typical historical betting data website. They also maintain a historical archive of starting price data and make it freely available online. This dataset also came with a few extra benefits. Some data which is gated by the premium tiers in the historical data files is made available freely, such as the volume. The dataset is also significantly larger than what was available from the historical pricing data. It

contains data on over 170,000 races in the UK going back as far as 2008, while the historical pricing data only goes back to 2016.

While the chosen dataset is excellent, it is worth noting that the market mechanism for the Betfair exchange is different than most markets surveyed in the literature. This is because Betfair exchange uses a CDA instead of a parimutuel system. The data from this set cannot be assumed to have the same characteristics as those found in the parimutuel markets.

3.3. Classification Design

Race classifications were chosen to represent various different clusters of races which appeared likely to impact the accuracy and efficiency of the market. Two groups of classifications were developed: external market factors and internal market factors. The external market factors involve information outside of the market but which may still ultimately be relevant. An equivalent in traditional investing would be grouping companies by industry. Internal market factors are those which can be determined simply by reading the current state of the market. Applied to traditional finance, this could be a metric such as PE ratio. Good factors need to have a large number of races per group and appear as though they may have some market impact.

For external factors, I chose race number, distance, hour of day, race type, location and number of runners. Race number and hour of day are chosen to reflect how the market might evolve over time. It is common for stock exchanges to have increased trading volume and activity at the start of the trading day and I imagine that similar habits may appear in racing. Distance, race type, location and number of runners can be considered largely analogous to the industry of a company. These represent different areas of interest for racegoers, owners and jockeys.

The internal factors chosen are pre-play trade volume and overround. Number of runners could also be considered to be a market factor in that it affects the number of selections to choose within an individual racing market. Trade volume can generally be considered a measurement of market popularity and it would be reasonable to guess that higher popularity results in greater market accuracy and efficiency. Overround is a more betting specific metric, which measures the total probability of one event occurring given the implied probabilities of the odds. Traditional bookmakers make money by offering odds with an overround greater than one as this means that a bet spread amongst all options can not be adjusted to make a guaranteed profit in any scenario. It would then seem likely that if any of the markets are offering an overround of less than one, a profit can surely be made.

3.4. Accuracy Metrics

A combination of different techniques are to be used. First of all, the existence or non-existence of the favourite-longshot bias must be determined. Secondly, accuracy test statistics are to be implemented to find the relative accuracy per grouping.

Methods for de-biasing odds are not used as these assume in the first place that the market is inaccurate. Instead I wanted to determine the level of market accuracy and not whether the odds could be transformed into something more accurate.

3.5. Efficiency Testing

Due to my choice of dataset, I do not have an opportunity to compare prices of exotic bets to parlays. This means that the most straightforward way to test for market efficiency is to subject the market to various betting strategies. Multiple different strategies are to be implemented

and compared to try and find a market inefficiency.

3.6. Software Architecture

The data will be broken up into three relational tables in order to increase operational efficiency. This is because the original dataset includes many duplicated sections of data, relevant only at specific levels of detail. The tables chosen are events, races and runners. Once the data is normalised this way, it becomes easy to perform operations on one section before joining it to another if it holds relevant details.

Test statistics and other details can be implemented fairly straightforwardly through Pandas. The key to this is making sure that categories can be easily interpreted and selected when calculating statistics.

Betting strategies are simplified as functions which simply choose which selections to bet on and how much. This means that details about profit and results can be handled in one separate instance for all different strategies. Betting strategies are easier to implement and alter as they do not have repeated code found within them.

4. Methodology

This section summarises the methodology used to obtain results. Further detail is explored in the implementation section.

4.1. Data Acquisition

Data is first acquired by downloading from the Betfair BSP price files directory. Once downloaded, they are then pre-processed into a more usable format. Win and place market files are stitched together, ensuring that no errors are caused in the process.

4.2. Accuracy

Implied and observed win rates are calculated per BSP group. These allow the shape of the distribution to be seen visually, as it should be linear. Each BSP bucket has a 95% Wilson confidence interval determined to see if the results are statistically significant.

A calibration regression test is done using logistic regression. This helps numerically identify favourite-longshot bias or other bias that may be present in the sample.

Brier and log-loss scores are determined per race originally and then averaged for final results. Skill scores are determined using the scores determined from the dataset as a whole as the baseline. This makes it clear whether a grouping can be considered more or less accurate than the dataset as a whole. Random samples of groups are taken to identify the mean skill scores and confidence intervals.

4.3. Efficiency

Three strategies are chosen to be compared. The first two strategies to be used are simple favourite-betting strategies. These strategies are chosen as they will very quickly identify one source of market efficiency, if present. They are also very common ‘‘system bets’’ used by punters at the racecourse. The first strategy simply bets one unit on every single favourite horse. An efficient market should not allow this strategy to make any profit, but it also should not result in a loss. The other favourite-betting strategy is the same, except the amount bet is scaled by the

implied odds of the horse winning. This strategy aims to avoid situations that may arise in the first strategy where an unlikely favourite wins and appears to result in a significant profit. The hedging strategy as described limits large wins from unlikely favourites.

I will also use a strategy adapted from Hausch et al. (1981). This strategy is chosen as it is the first major example of a betting strategy being used to demonstrate market inefficiency in horse racing. As the original strategy is used in the context of a parimutuel market, the impact of a bet on the odds price is known beforehand. This is not possible with my dataset (though it is possible while betting live) and so I have altered the bet sizing portion of the strategy to use the Kelly Criterion. That is: $f* = (bp - q)/b$ where $f*$ is the fraction of wealth to bet, p is the likelihood of the bet winning, q is $1 - p$ and b is the proportion of wealth won if successful (Thorp, 2006). This means that unlike the original strategy, it is not suitable for use in real betting environments as it does not take into account how the bet will move the market. It is still however useful for identifying market inefficiency, but the profits reported may be exaggerated from what is truly possible.

For the more advanced strategy, I chose to place a bet when it is determined that the expected edge is greater than 25%. This means that a significant amount of confidence is required and if the strategy is good, it should in theory result in a profit of 25% every time a bet is placed. The bets use a scaled Kelly criterion of 10% in order to avoid periods of boom and bust.

The strategies are tested against random samples of data in order to bootstrap their expected return and confidence intervals.

5. Implementation

In this chapter, I will discuss the process for developing the software used to come to my results.

I decided to use Python through Jupyter Lab to write and run my software. Jupyter Lab is the standard for data science projects using python. Python features extensive libraries for data analysis and statistics. Particularly useful in this project is the Pandas library, which provides a significant amount of functionality for working with structured data.

To plot my results, I used the program Gnuplot. Gnuplot is available as a Jupyter kernel and is capable of reading data from files. It also natively supports a variety of output formats, including PostScript which is easily embeddable within this document. Separating data generation and data visualisation also makes things simpler, especially when one data file is used to create more than one plot.

For version control, I used Git to track changes to the project over time. This included changes to the report itself as it is written in a text file format and then processed to create a PDF. The repository is mirrored to GitHub, backing up my work and ensuring that a lost or stolen device could not cause me to lose my work.

5.1. Data Acquisition

I produced a Python script to automatically download all of the files found in my dataset and processed them to be used within Pandas. The script saves the data to binary files for quick retrieval. It makes sure to not re-download files it has already obtained so it can easily update the existing database. As well as this, it automatically slows down when rate limited to ensure that all files are downloaded. The total size of downloaded files is over 600MB.

After downloading, the data is preprocessed into a format that makes it easier to use for my purposes. This involves transforming the flat file structure into a relational one, separating unrelated data. Entries for some columns need altering. Locations need to be changed as the format for location changes at one point in the files, moving from truncated names to full location names. Some selection names erroneously have their runner number included and so this is removed. Distances are given as a string format containing miles and furlongs and so are transformed to a single integer representing the number of furlongs.

Win and place data are in their own separate files originally and so must be joined together in order to use both sets of data. The script identifies matching entries in both sets of files using the race time and selection ID which is a unique pairing. Pandas is able to validate this as a one-to-one pair when using its join function. This enables the place odds and win/lose result to be included within the dataset.

The result of this processing is 3 binary data files totalling just under 100MB using the structure found in tables 1, 2 and 3.

Field	Data Type	Notes
`event_id`	`Integer`	Unique identifier provided by Betfair
`location`	`String`	
`event_date`	`Datetime`	

Table 1: Events Structure

Field	Data Type	Notes
`race_id`	`Integer`	Unique identifier provided by Betfair
`race_type`	`String`	Indicates the format of the race: handicap, hurdles, chase, etc.
`place_n`	`Integer`	Number of places for this race.
`race_time`	`Datetime`	
`distance_f`	`Integer`	Furlongs are one eigth of a mile long.
`race_no`	`Integer`	Races are given numbers indicating their sequence position within an event.
`event_id`	`Integer`	Relational key.

Table 2: Races Structure

Field	Data Type	Notes
selection_id	Integer	This identifier is unique per horse, not per entry.
selection_name	String	
win_lose	Boolean	Win=True, Lose=False
place_win_lose	Boolean	
bsp	Float64	Betfair Starting Price
place_bsp	Float64	
implied_odds	Float64	1/BSP
ppwap	Float32	Pre-play weighted average price.
morningwap	Float32	Weighted average price of bets before 11AM GMT
ppmax	Float32	Pre-play max price.
ppmin	Float32	Pre-play min price.
ipmax	Float32	In-play max price.
ipmin	Float32	In-play min price.
morningtradedvol	Float32	Trade volume before 11AM
pptradedvol	Float32	Trade volume before the off
iptradedvol	Float32	Trade volume after the off
race_id	Integer	Relational key.

Table 3: Runners Structure

5.2. Pre-Processing

I pre-processed my data to include some more fields that would be useful. Pre-processing fields like this means that the calculations do not need to be repeated later on. This is very important and saves a significant amount of processing time for a dataset as large as this.

The first field to be added was a normalised probability field. This is useful as the implied odds given by the starting price (almost) never sum to be one. The sum of implied odds given is known as the ‘‘overround’’. An overround greater than one for a back market shows that there is negative arbitrage and a weighted selection of back bets is guaranteed to lose money. Since the data for this project is using the BSP, it can trivially be shown that it is possible to arbitrage almost every single market. This is because for BSP, both back bettors and lay bettors are given the same price. In real terms this arbitrage is actually impossible to achieve however, as the BSP is not possible to know before the off, and by betting this way you are certain to move the market.

I then added fields for partial pre calculation of the Brier score and Log-Loss. This is particularly useful as it means that to determine the scores for a certain subset of races, they can simply be summed using the groups generated in the next section. Logit values are calculated to be used in the logistic regression.

Returns of a single unit bet are also pre-calculated. The return is simply 0 if the runner did not win or the BSP if it did. This is useful for the testing of betting strategies later. These returns can simply be scaled by bet amount to find the returns of different sized bets.

5.3. Categorisation

All categories are assigned as columns to the data so that they can be simply referred to by name later on. This makes it easy to group the data appropriately using the groupby or pivot_table functions. A list of categories holds the name of every category as well as a filter for which groups should be observed. This filter is necessary to remove small groups that do not provide a lot of data or to make it easier to analyse categories with a particularly large number of groupings.

The groupby and pivot_table functions make it simple to perform aggregate calculations per group defined this way. Groupby enables a function to be performed for every group of the category, such as summing the returns of bets. Similarly pivot_table enables this but with two different grouping functions. This is useful when groups need to perform calculations over a range of values, such as date or BSP.

5.4. Mean Returns by Implied Probability

Runners are grouped by BSP values and the mean win rate for each value is determined. BSP values given by the dataset are continuous instead of being representative of the tick sizes available using the exchange and so they are first calibrated to align with the closest actual trade value possible. For each different BSP value, the mean win rate, the number of observations and the total trade volume are recorded. When graphed, the tick size is taken into account so that wider groups are not misrepresented as being more popular. A Wilson confidence interval is calculated for statistical significance testing.

Logistical regression is performed on the entire dataset and per grouping to determine the slope and intercept of the plot. Confidence intervals for these values are also generated for comparison.

5.5. Test Statistics

To calculate the Brier and log-loss scores, runners do not need to be grouped in two different ways. The scores can simply be calculated as an aggregate of the whole dataset and per category grouping. Each group has races sampled 1000 times to determine the expected scores as well as their 95% confidence intervals.

5.6. Trading Strategies

Trading strategies are implemented using a system in which the strategy function receives a list of available runners to bet on and returns a ‘‘bet slip’’. The bet slip contains information about which race and runner it chose, as well as the amount bet and whether or not the selection won. The amount bet is able to be submitted as either an absolute value, or as a percentage of the currently held bank value.

The bet slips are passed to a function which actions the bets and determines the PNL for each one. It also sorts the bets over time to generate a timeline of events.

Each betting strategy is written using vector based functions. This allows the underlying library (Pandas or Numpy) to perform many operations in parallel. There are two primary benefits to this. Firstly, it significantly improves the readability of the code as the need for looping is removed. The code is much more representative of what is actually happening. Secondly, there is a major performance increase by writing the software in this way.

The simple strategies naturally have simple implementations. Every favourite is identified (this is actually pre-calculated to save time) and bet on. The size of the bet is either a constant single unit, or for the case of the hedging strategy, the inverse of the BSP.

The more advanced strategy has a more complex implementation. First of all, all races without 2 or 3 places are discarded as determining the Harville odds for any more than 3 places becomes significantly more complex. This is not a major issue as less than 1% of races in the data pay out more than 3 places. The place odds are determined by using the Harville method to produce the odds of the horse placing second or third and adding this to the probability of the horse winning. The Kelly growth criterion is then calculated to determine the percentage of current wealth to bet and the bet slip is produced.

For the purposes of analysis, bets generated by the strategies are sampled 1000 times. This enables determination of the expected returns of the strategies including confidence intervals. If this is not done, the results of strategies run over the data are simply individual results and are not comparable.

6. Results, Analysis & Evaluation

6.1. Initial Observations

For most of the groupings, trade volume follows the number of races as can be seen in Figure 1. This is of course different for the categorisations of volume (per race) and overround, which have been specifically broken up into 10 sections of equal size.

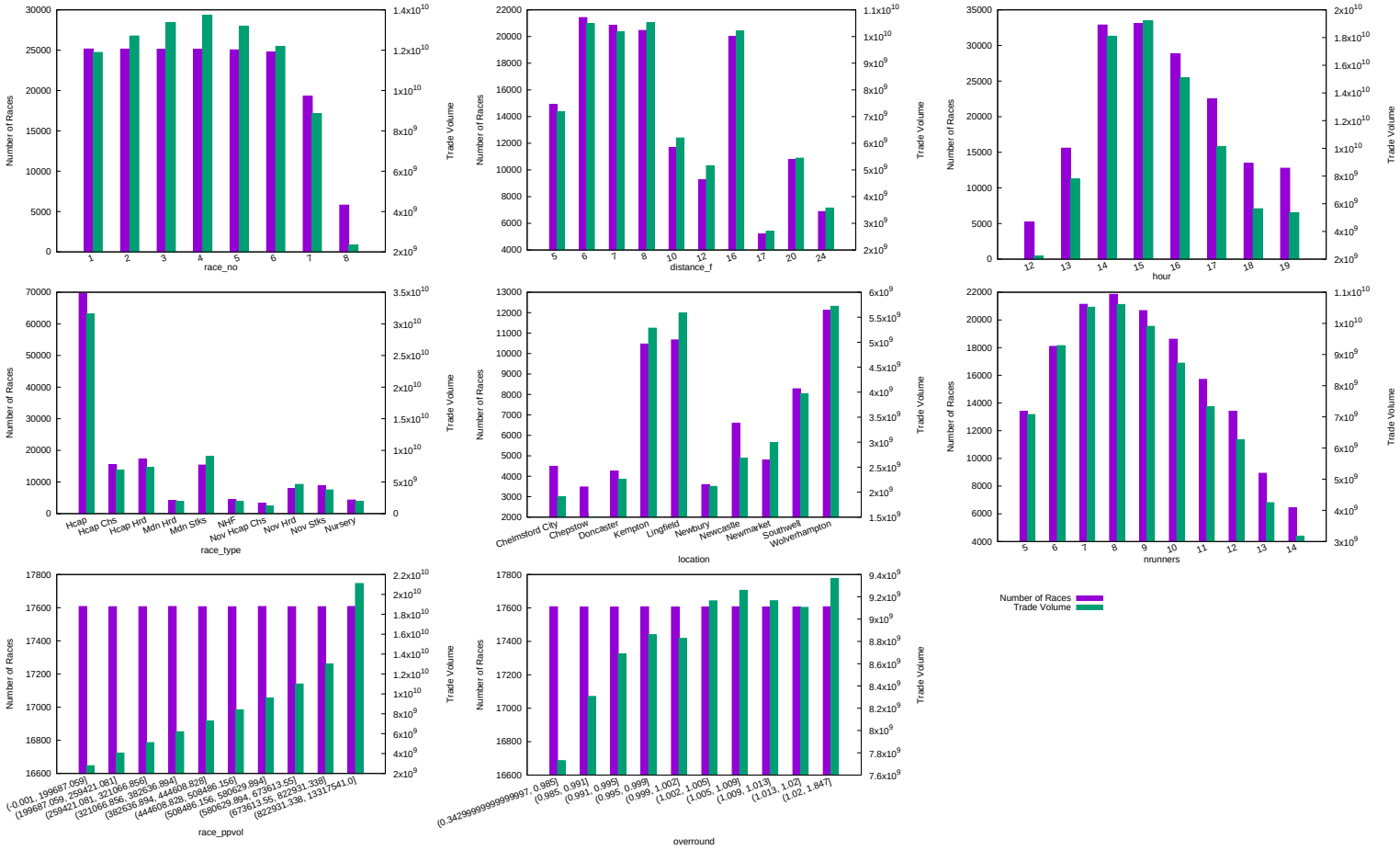


Figure 1: Number of Races and Trade Volume per Category

For the entire dataset, the volume traded appears to be roughly normally distributed about the mean implied probability of 0.5. The number of selections however is positively skew, with significantly more selections being given a lower implied chance of winning. This shows that most horses will be considered unlikely to win a race. More money is traded on a smaller proportion of horses with greater chances of winning.

An interesting observation is the somewhat frequent spikes in prices. These are all observed at the points in which the market tick size changes (BetexTrader, n.d.). The tick size is the difference between prices that are allowed to be offered. I have used the tick size to group selections based on their BSP, as the reported BSP is not the same as the actual price matched against. The tick sizes are used to adjust the frequency values in the plots.

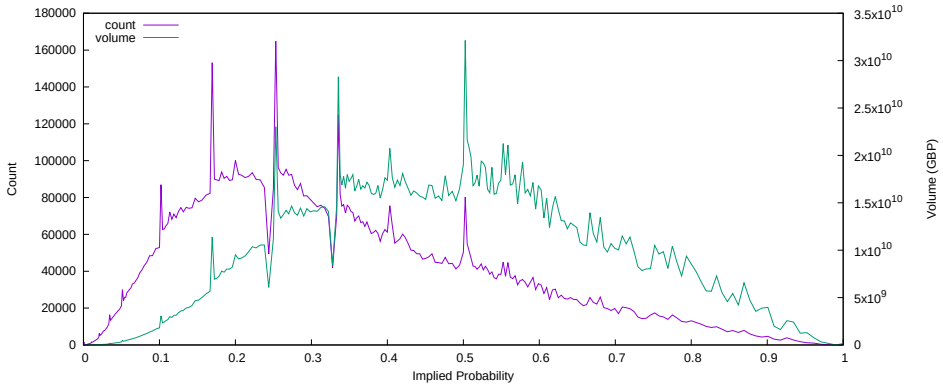


Figure 2: Frequency of Count and Volume of BSP Values Across All Races

The volumes per grouping tell a largely similar story. For each different grouping, the distributions are typically the same as per the whole dataset, except scaled by quantity. Distance, hour, location, and number of runners all fit this trend.

Race number becomes more positively skew as it increases. This shows that over the course of the racing day, bettors are more likely to trade on horses with a lower implied chance of winning. We can see this is not simply due to the odds being offered as the frequency plot remains almost exactly the same across all races.

The volume for race type is another outlier. Handicap races are massively positively skew while others tend to be negatively skew. Handicap races are by far the largest race type in the data set and so have a significant impact on the overall weighting. It could be the case that on a dataset without featuring handicap markets, volumes generally tend to be negatively skew, instead of symmetric.

For both total pre-play trade volume and overround, the volume of the market tends to shift towards being negatively skew as they increase. This trend is particularly pronounced for pre-play trade volume. As total trade volume increases, trade tends to centre around horses with a greater chance of winning. It is so extreme that in the highest bracket of volume, well over 50% of trades are being made on selections with an implied probability of winning of over 50%. It could be speculated that this is signifying an increase in professional trading activity, where traders try to scalp prices on popular horses for bigger races. This would require more research to validate.

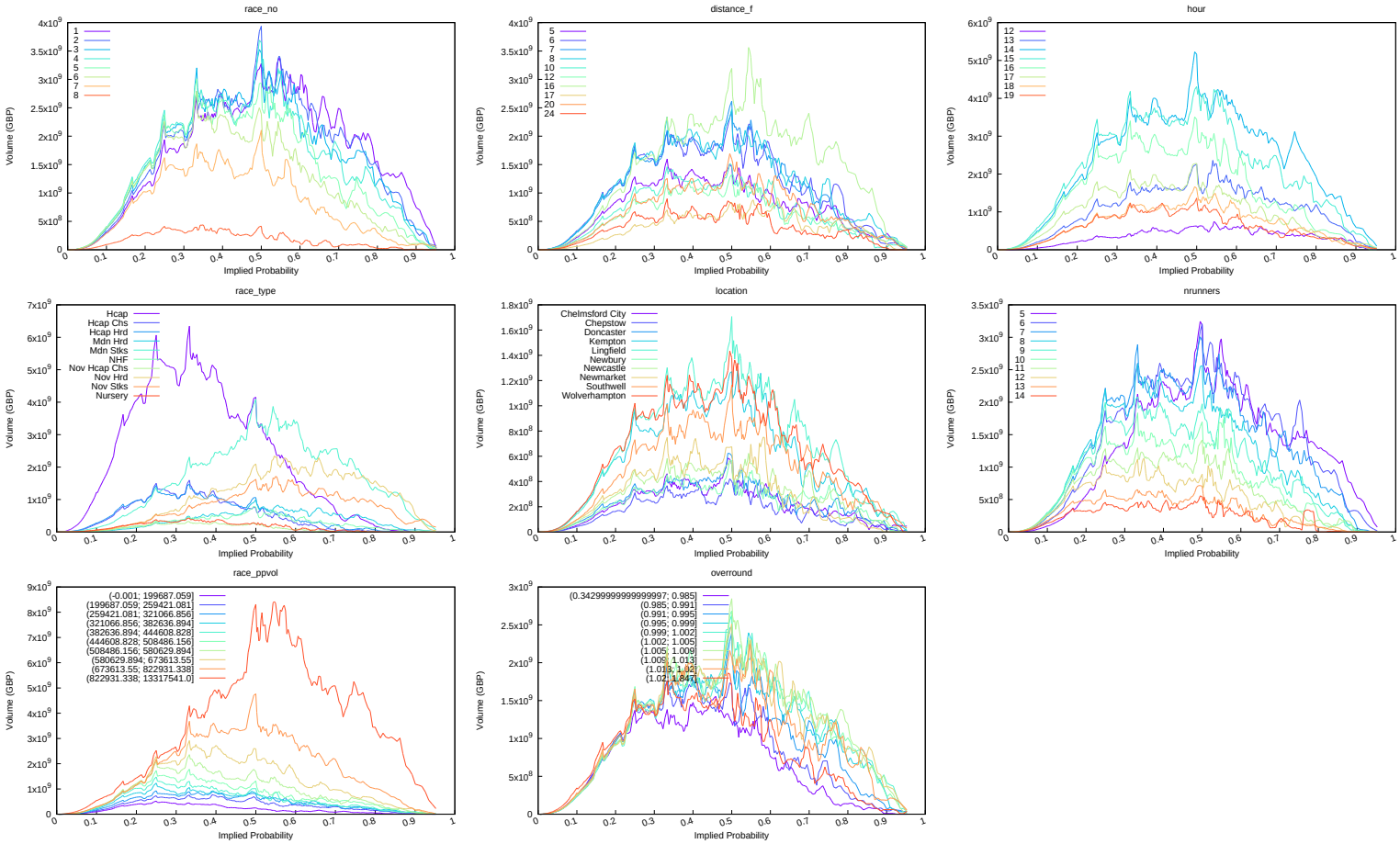


Figure 3: Frequency of Volume for BSP Values by Category

Frequency follows the same trend for all categories except for pre-play trade volume. All other groups are positively skew, being scaled relative to their group size. For trade volume however, the frequency becomes more negatively skew as the total volume increases. This is similar to the previous observation for trade volume. As pre-play trade volume increases, the distribution of probabilities assigned to horses changes as well. Prices going down is indicative of greater market confidence in the favourite horse.

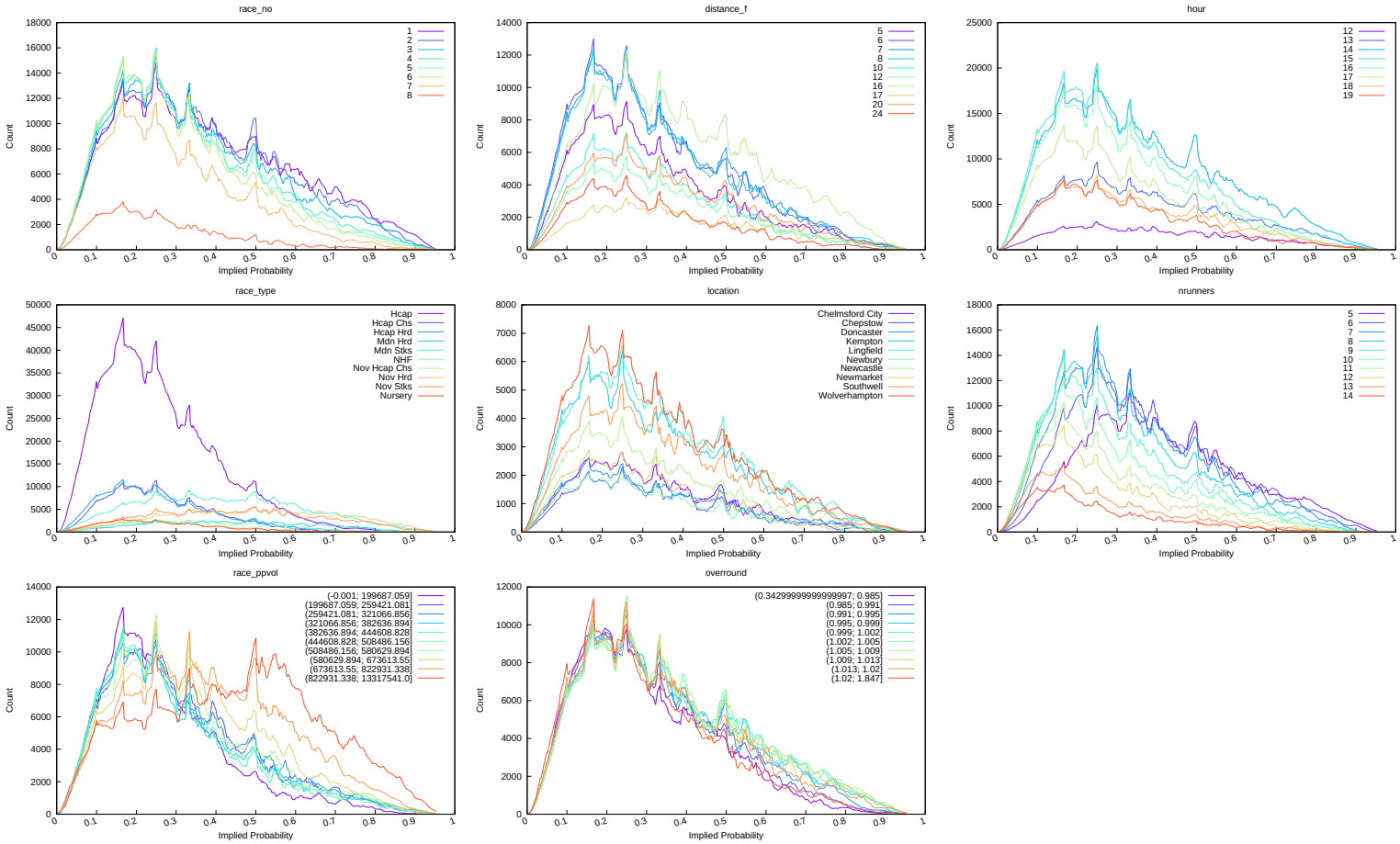


Figure 4: Frequency of Count for BSP Values by Category

6.2. Accuracy

6.2.1. Favourite-Longshot Bias

If the favourite-longshot bias is present within this sample, I would expect to find that favourites win more often than implied by their given betting odds. Similarly, longshots should win less often than implied by their betting odds. From research done in American markets (Griffith, 1949; Ali, 1977), I am expecting to find the favourite-longshot bias within my sample.

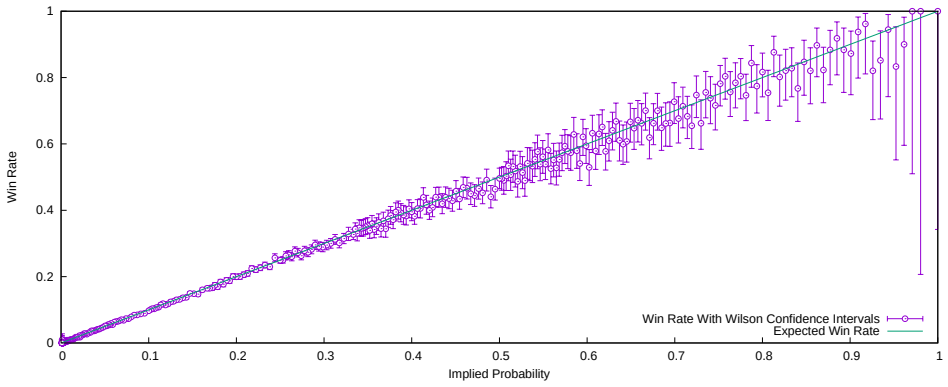


Figure 5: Win Rate Compared to BSP For All Races

When plotting expected win rate against true win rate, it appears as though the favourite-longshot bias is not present within the entire sample. Figure 5 shows this using Wilson confidence intervals. If the favourite-longshot bias was present, the mean win rate would appear greater than the expected win rate for low implied probabilities and lesser than for high implied probabilities. This would result in an "S" shaped plot. Almost every confidence interval contains the expected win rate and so this is the first indication the favourite-longshot bias is not present.

A logit regression is employed to determine sample calibration. If the sample is perfectly calibrated, then the slope should be one and the intercept zero. The results of the regression are in Table 0.

Parameter	Estimate	95% Confidence Interval	Interpretation
Intercept	-0.0064	(-0.0169, 0.0042)	No overall bias
Slope	0.9967	(0.9913, 1.0022)	No probability distortion

Table 4: Logistic Regression of Implied Probabilities Against Win Rate

The logistic calibration model finds no evidence of systematic miscalibration within the sample. Both the intercept and slope values are not significantly different from what should be expected. This indicates that the market values accurately reflect the outcomes and so there is no favourite-longshot bias. The logistic regression is calculated for every grouping of races.

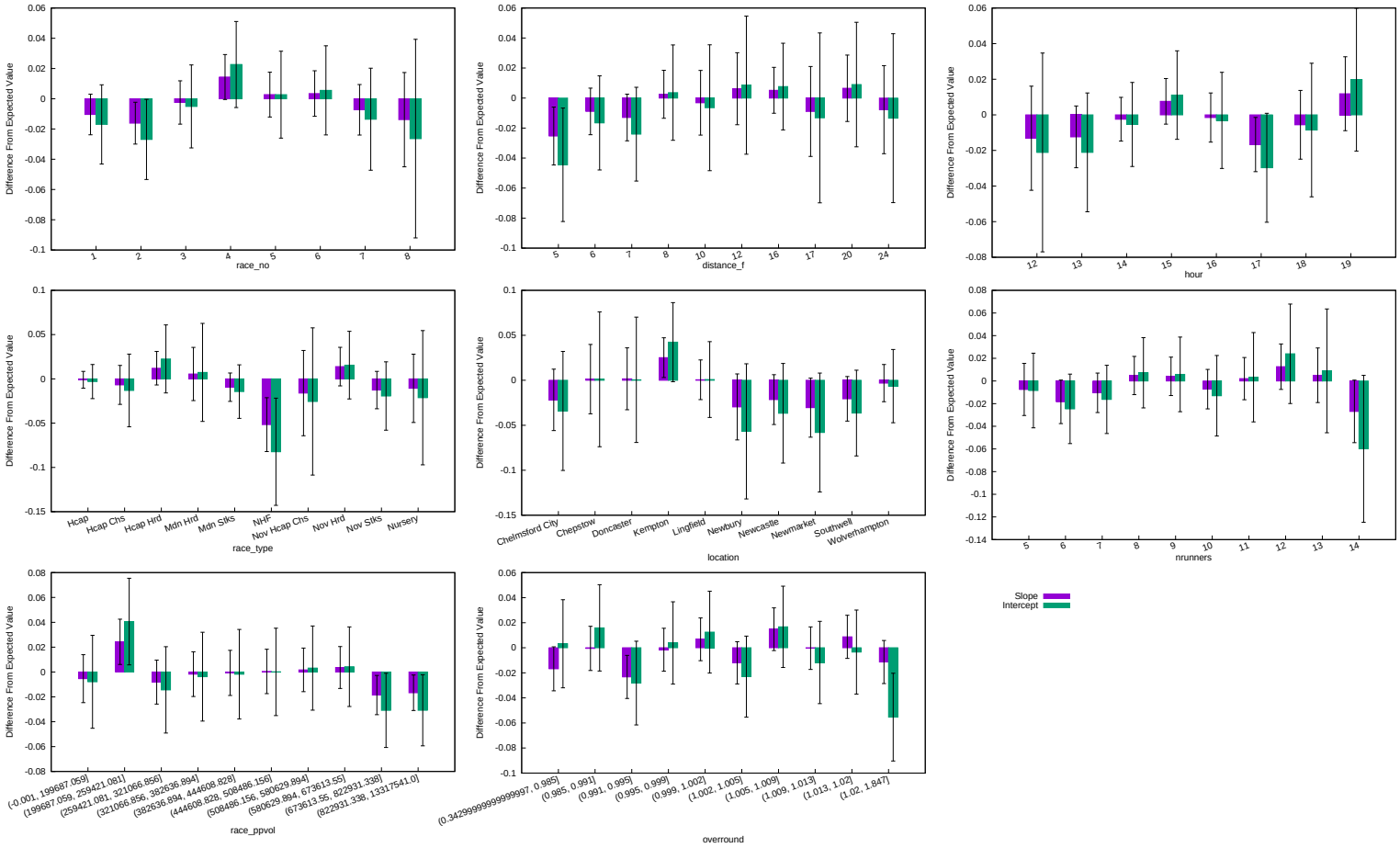


Figure 6: Error of Expected Values of Logistic Regression

Almost all groups appear to exhibit no significant bias or probability distortion. There are a few exceptions. Most of the exceptions are minor, being just outside of the 95% confidence interval. These are race number 2, hour 17 (5PM), and the extreme trade volume values. All of these have one or both of the slope and intercept significantly lower than their expected values. A slope value less than one is what would be expected of a sample featuring the favourite-longshot bias as this shows an undervaluing of favourites and an overvaluing of longshots. An intercept less than zero demonstrates systematic overestimation of winning probabilities. These values could suggest that the favourite-longshot bias is present, but are not conclusive.

Bigger variation is found in the highest overround group, 5 furlongs and the NHF (National Hunt Flat) race type. The overround group features a significantly lower intercept and the NHF and 5f groups have a significantly lower intercept and slope. Because the slope is not significantly lower than 1 for the highest overound group, this is not consistent with the favourite-longshot bias. Instead, this is an overpricing of all horses.The NHF and 5f groups are consistent with general overpricing and what would be expected of the favourite-longshot bias.

These few counter examples are consistent with what should be expected for false positives from this sample. With a 95% confidence rate and 76 different groupings being identified, 3.8 false positives are expected.

This is a stark contrast from most previous research, with the only other example I know of a racing market not exhibiting this bias being the Hong Kong Jockey Club (a parimutuel market) (Busche, 1988). I expect the primary reason for this difference to be the use of a CDA for price discovery, rather than a parimutuel system. The CDA encourages greater price discovery and its format is familiar to those with experience in financial markets. This could suggest that the composition of the Betfair Exchange market leans further towards groups of insiders or professional bettors, rather than normal punters. Further investigation would be required in order to determine this.

6.2.2. Forecast Test Statistics

To determine the test statistics, the statistic is determined first for each individual race and then averaged across the whole grouping. This is done as we are less interested in the score as determined per runner as we are per race. This is because each runner is not an isolated event, and is instead packaged within a race. Organising by race prevents extra external factors from affecting the results. For the data set as a whole, the Brier score is determined to be 0.0946 and the log-loss is 0.3157. Skill scores can be used to show that both of these metrics are better than randomly guessing the likelihood of a horse winning a race.

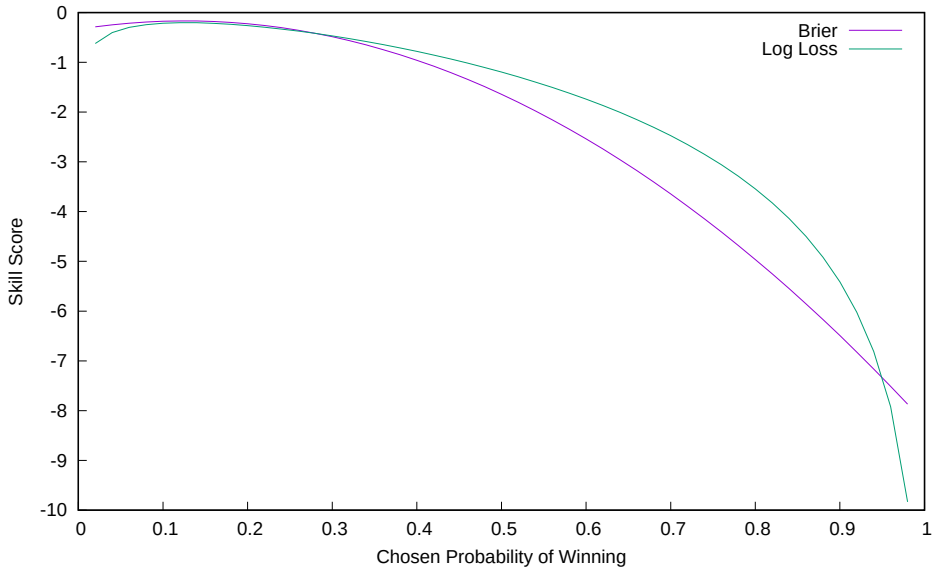


Figure 7: Skill Scores of Random Predictions Compared to Market Predictions

Figure 7 compares different rates of estimating the probability of a horse winning. At 1, the model predicts every horse to win every race with a 100% chance and at 0 it predicts every horse to not win with a 100% chance. The comparison makes the differences between Brier and log-loss extremely clear, as log-loss drops sharply at higher estimated probabilities due to its greater punishment of incorrect guesses. Both of these scores show that the market performs better than random.

I would expect the skill scores to become greater for groups of larger sizes. This is because the larger pool would provide more data for bettors to use to make predictions with. The sizes of the groups can be found in Figure 8. It would also make sense for some particularly small groups to have outlying skill scores as they will be more susceptible to variance.

To identify accuracy within the different groupings, brier and log-loss are calculated the same way as before, accounting for selections within races.

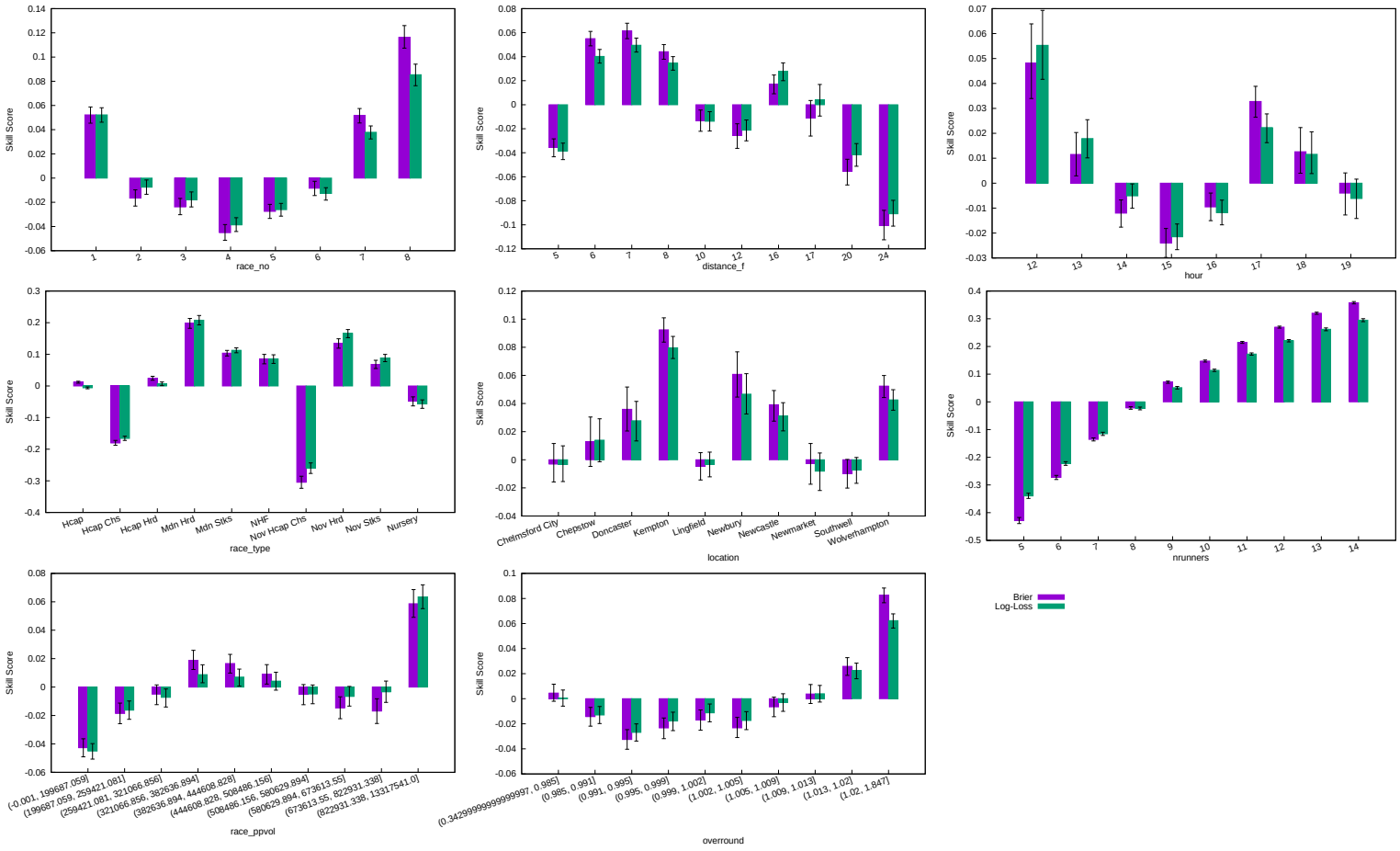


Figure 8: Brier and Log-loss Skill Scores by Category

Figure 8 shows the Brier and log-loss skill scores with 95% confidence intervals. It demonstrates that both skill scores are largely similar. There are a few occasions where the scores conflict and one is negative while the other positive. In all instances of this, the skill scores are both very close to zero. The difference will be due to slight differences in the distribution of skill scores as seen in Figure 6.

Some skill scores appear to be correlated with the size of the categories or the total trade volume. (Figure 1). Categories that do appear correlated are: race number, distance, hour, and location. Race number appears to have the skill scores negatively correlated with volume, as does hour. However, distance and location have skill scores that positively correlate with volume. The contrast between these categories shows that volume can not necessarily be used as an indicator of market accuracy.

Other categories appear to have skill scores unrelated to the relative volume or race count.The prime example is number of runners; skill scores increase as the number of runners does. This behaviour is to be expected because of our test statistics. A greater number of selections in a race means that the average probability estimate will decrease and the number of non winning horses increases. Number of runners has the greatest range of skill score values and

appears to be the most significant factor in determining the accuracy of the markets by these metrics.

Race trade volume and overround both exhibit a behaviour of having low or negative skill score before sharply rising at the high end of the categories. Overround is a representation of the total probability of prices given in the market. For an overround of 1, all prices add up to a total probability of one event occurring being 100%. Odds offered by traditional bookmakers typically have an overround much greater than one, this is the margin that they make. Because of this, I would have expected the accuracy for overround to peak around 1, as at higher and lower values there are more prices which are explicitly wrong. The results here contradict that expectation. Trade volume per race is mostly a measure of race popularity, and so increased accuracy for the greater trade volumes makes sense. There is more money in the market and probably a greater proportion of informed bettors.

The only category not mentioned thus far is race type. There seems to be no relationship between the race type and accuracy. Size and volume of the groups does not impact the accuracy here, where handicap races (by and far the most popular) have a similar score to handicap hurdles or nursery races.

6.3. Efficiency

Three betting strategies were trialled in an attempt to exploit inefficiency within the market. The first two strategies bet on the favourite horse for every race. The first one simply bets one unit on every favourite, while the second bets the inverse of the starting price. The second strategy makes a weak attempt to ‘‘hedge’’ by betting lower when it is apparent that the probability of winning the bet is lower. The third strategy is adapted from the technique used by Hausch et al. (1981).

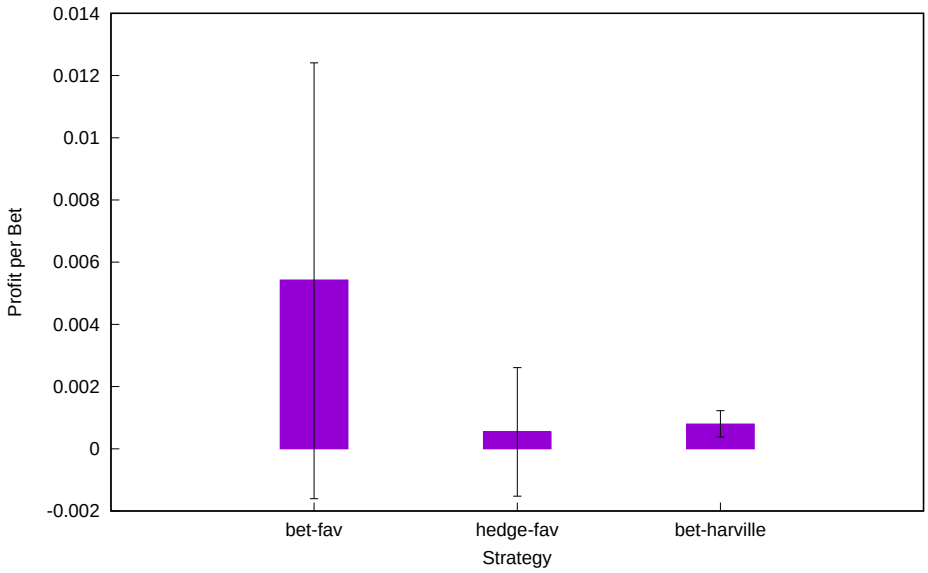


Figure 9: Results of Betting Strategies Applied to All Races

The results of these strategies when applied to all races can be seen in Figure 9. Both the favourite betting strategy and the favourite hedging strategy do not have mean returns statistically significant from zero. This is to be expected in an efficient market as the expected value of the

profit of a bet should be zero.

The more advanced strategy appears to generate a consistent profit per bet. Profit is greater than zero within the 95% confidence interval. This demonstrates a theoretical lack of efficiency in the market. It is worth remembering however that this strategy does not take into account price movements caused by the placement of large bets. This was an important feature of the original strategy that I have not been able to replicate for this task. If the market behaved as if there was infinite liquidity at quoted BSP prices, this strategy would be effective at generating a consistent profit.

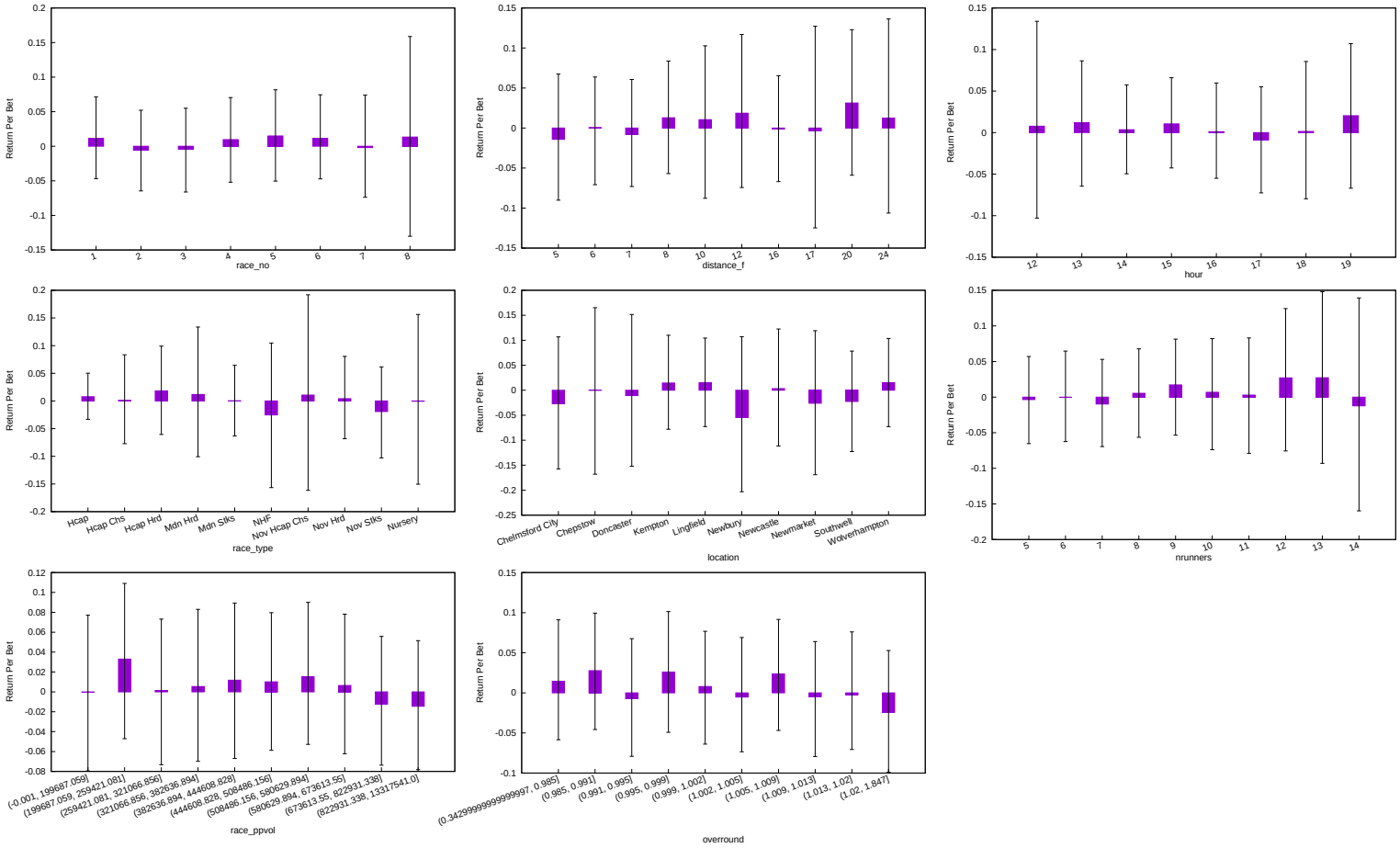


Figure 10: Favourite Betting Strategy by Category

When applying the two basic strategies to the different categorisations, no group appears to perform particularly well or poorly. In Figure 10 the results for the favourite betting strategy are shown. Unlike the accuracy test statistics, no trends appear to emerge. None of the groupings have returns significantly different from zero.

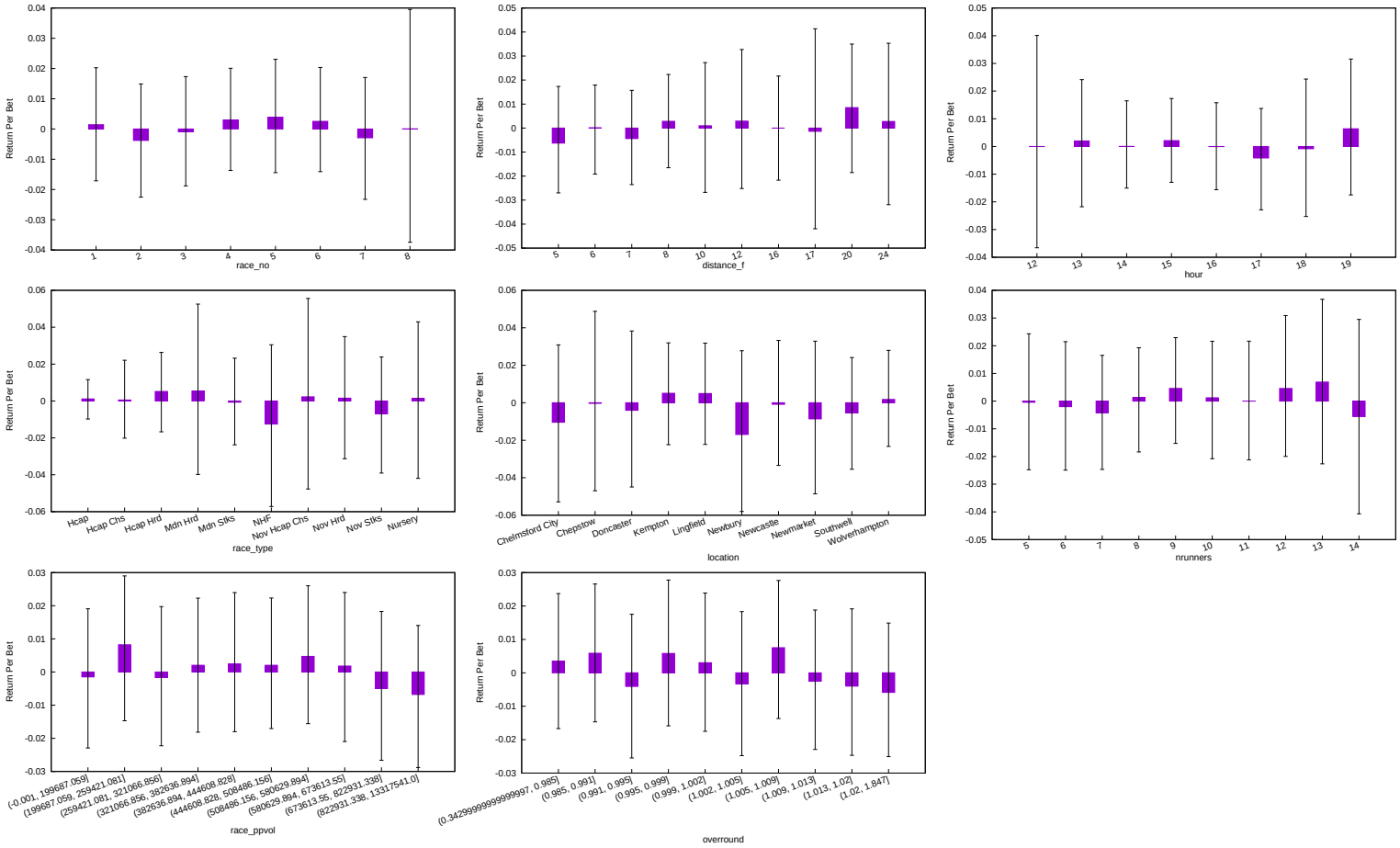


Figure 11: Favourite Hedging Strategy by Category

The favourite hedging strategy is the same as the favourite betting strategy in that no patterns can be found in the results. The returns are also not significantly different from zero.

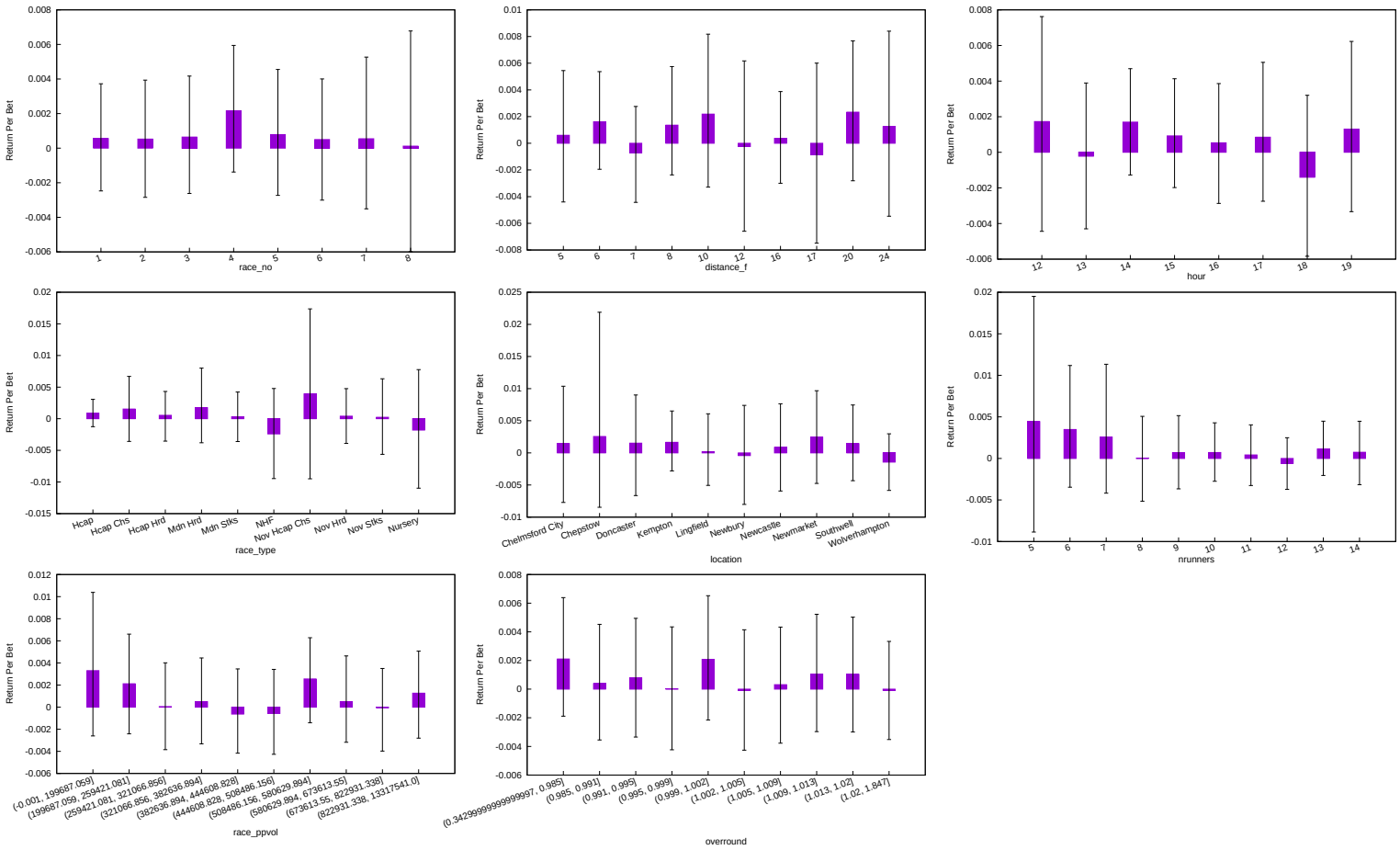


Figure 12: Advanced Betting Strategy by Category

Unlike the results for the entire dataset, no grouping has a return significantly different from zero with the advanced strategy. This is likely explained by the groupings being much smaller than the whole sample, and so the confidence intervals are wider.

It cannot be said that for any of the different groupings an inefficiency has been found, though it could potentially be said of the entire dataset.

7. Legal, Social, Ethical & Professional Issues

7.1. Intellectual Property

To perform this research, I made use of data provided by Betfair. The page serving this data provides no information regarding its use. There is a reference to the directory on one FAQ page (Betfair, n.d.), which states that the data is available to download.

7.2. Incentives & Insider Trading

The availability of prediction markets for certain events can incentivise negative acts that otherwise would not be undertaken. At the Super Bowl in 2021, a streaker reportedly bet $50,000 that there would be a streaker (O'Reilly, 2021). The availability of perceived easy money from this bet directly encouraged a man to engage in illegal behaviour. This was even rationalised

(somewhat ironically) as an arbitrage opportunity. Since the cost of streaking is fixed (ticket costs, transport and fines), any large enough bet placed will result in a guaranteed profit assuming the bettor is able to successfully streak.

In theory, a prediction market could be engineered in such a way as to discourage this. If information was efficiently priced into the market, the possibility of someone going out of their way to streak for market gain would be reflected appropriately in the price. As with all prediction markets, money talks and a large bet on an action signals strong insider information to the market. It is likely that markets for betting on the possibility of a streaker are now more aware of this possibility and are making sure to price it in.

Prediction markets do not just incentivise bettors to take actions that impact the outcome of events. They also incentivise insiders to bet large amounts on outcomes they know beforehand. For the purpose of the market, this may be considered good as it pushes the market towards greater accuracy. In many scenarios however, this information is confidential or sensitive. A U.S. soldier was charged with using classified information about the capture of Nicolás Maduro in Venezuela (U.S. Department of Justice, Office of Public Affairs, 2026) to bet on outcomes of the military operation using Polymarket. In the past, sale of state secrets would have to be done face to face with a foreign agent. Now instead, soldiers are incentivised to sell out their country publicly.

Attempts to fix horse races are taken very seriously by the BHA. A 2023 conspiracy to fix a single race by stopping a horse from achieving its best possible place resulted in a 9 year disqualification for the jockey and a 12 year exclusion order for the owner (British Horseracing Authority Judicial Panel, 2025). Investigations and punishments like these enable horse racing markets to operate in an environment that is generally believed to be free from corruption. This is however relevant to a very small portion of highly regulated prediction markets within the UK. When prediction markets are expanded to include all manner of events, specialised investigation boards are impossible to create. It becomes the job of the courts to regulate behaviour, rather than the industry being able to self-regulate.

7.3. Gambling

The gambling industry profits from habitual gamblers with addictive personalities. For many people, gambling can become a significant problem. Most gamblers are essentially guaranteed to lose money in the long run, causing financial harm if not managed properly. A bookkeeper fundamentally makes money by preying on peoples' hopes of winning. By engaging with gambling markets, you help to enable problem gambling. Gambling markets are a zero sum game, to succeed you must take money from someone else. By implementing a successful strategy, you are more than likely directly taking money from gamblers with no thought-out system or genuine racing model.

Gambling markets look bad when framed this way, but fundamentally they operate extremely similarly to traditional financial markets. Huge amounts of people use the stock market regularly to invest in companies they believe will be successful, but the process is essentially the same: a gamble. The primary difference is the social framing of the two. Big established companies and investors are seen as sensible and reasonable while gamblers are considered unreasonable. Of course, the engagement that people have with these two markets is very different, even if they operate similarly. When investing in stocks and shares, most people typically invest in a fund curated by an institutional investor. This is not guaranteed to increase in value by any

means, but historically has proven to be successful for most average people. If an investor begins to choose individual stocks, then they become less likely to succeed. This is equivalent to typical gambling practices, where the bettor chooses each individual horse to bet on.

7.4. Animal Welfare

Horse races and some other popular prediction markets (such as greyhound racing) rely on the use of live animals. In some cases the animals may become seriously injured or die. Steps have been taken to reduce this number over time. From 2009 to 2025, incidents of race fatality per 1,000 starts fell from 2.00 to 1.07 in the US (The Jockey Club, 2026). In the UK, Aintree racecourse announced in 2013 significant changes to the prestigious Grand National race in the name of improving safety (Scargill, 2023). The maximum number of runners was decreased from 40 to 34. As well as this, multiple course features were altered including the lowering of some fences.

Because of the associated danger, some animal rights groups have attempted to improve conditions for animals or outright ban the events from taking place at all. Any changes imposed may have significant impacts on the markets, as models need to be recalibrated to account for the different environment. Additionally, the availability of the market could be completely jeopardised if events are banned.

8. Conclusion

There has been a surge in popularity of prediction markets over the last few years as services like Polymarket and Kalshi have become available. The recent legalisation of online sports betting in the US has also contributed. Horse racing is a particularly popular example of a prediction market and has a large amount of readily available data for similar repeated events. Results uncovered from racing market data can be applied to wider financial markets as a whole, making them valuable research targets.

This report builds on prior research in horse racing markets. It surveys the literature for determination of accuracy and efficiency in those markets. Novel research is contributed by first of all looking at a CDA market instead of a parimutuel market (which has typically been the focus of prior research) and secondly by decomposing the market into distinct sub-groups to observe.Beyond simply determining whether the Betfair exchange can be considered accurate or efficient, I have attempted to find the relative contributions of different groups of races to this accuracy and efficiency.

Accuracy was measured in two ways. The first method compared betting prices to the expected win rates implied by the odds. Unlike existing research in the American markets (Griffith, 1949; Ali, 1977; Kanto, 1992), I found the implied odds to closely match the true observed odds. For the whole dataset and when decomposed, there was not any favourite-longshot bias. It appears to be largely accurate. The second measure of accuracy was two skill scores: Brier and log-loss. Initial measurements for the entire dataset demonstrated that the market was more accurate than randomly guessing winners. When decomposing into different categorisations of markets, it was found that the number of runners in a race was the most significant factor for market accuracy.

In an attempt to measure efficiency, three different betting strategies were implemented. Two of them were naive favourite-betting strategies and the third was a method adapted from Hausch et al. (1981) which attempts to derive odds for a horse placing from its win price. In

almost all circumstances, the strategies were not able to produce returns significantly different from zero. The one exception is when the advanced strategy was applied to the dataset as a whole. This potentially indicates some level of market inefficiency, but the strategy is not robust enough to confidently claim that for now.

8.1. Future Work

To further the discovery of market accuracy, it would be beneficial to compare the use of other forecasting methods to the betting odds. Other methods could include polling and model-based forecasting. This would help to establish whether the implied odds given by the markets are actually the best forecast available. Other forecasting models may also help to establish a lack of market efficiency. If another forecast is able to provide better odds estimates than the market then a strategy could be developed similar to the one used in this report but with the market win odds substituted for the forecast odds. This would likely be identifying a different form of efficiency however, as information from outside historical prices would be included in the strategy.

This report focusses solely on BSP data from the Betfair exchange, but Betfair also runs its own sportsbook and tote betting facilities. It may be beneficial to compare the results found here to data obtained from those alternate markets. That would enable a direct comparison of market types as most existing research has been done in the US and on a different set of races.

It would be worth exploring different parameters for the strategies used or different strategies altogether. While I did experiment with different settings for the strategies, I have only presented the state I determined to be the best performing through manual testing. It may be possible that through a more thorough examination of the different parameters available, the strategy would succeed in demonstrating a lack of place market efficiency. Additionally, improvements on the strategy I adapted have been published (Hausch, 1985; Lo, 1995) and other strategies have been developed (Bolton, 1986). Other strategies may prove successful in demonstrating a lack of efficiency in the market.

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